Crisis-Aware AI Forecasting Final Report

Crisis-Aware AI Forecasting for U.S. Equity Indices: Evidence from the S&P 500 and Nasdaq-100

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Application-Oriented Python-Based Independent Research Project

Research Area: Financial Machine Learning; Macro-Finance; Crisis Regime Forecasting; U.S. Equity Index Prediction

Primary Methods: Long-History Panel Construction; Machine Learning; Ensemble Learning; Conformal Prediction; Backtesting; Mechanism Validation

Project Coverage: Day1-Day27 empirical core; Day28-Day36 paper-packaging and final-audit layer

Interpretation: Predictive and mechanism-consistent evidence; not financial advice, not causal proof

Final Corrected Quantitative Version

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Abstract

This final report summarizes an application-oriented Python-based independent research project that develops a long-history crisis-aware AI forecasting system for U.S. equity indices. The project focuses on the S&P 500 and Nasdaq-100, combining macro-financial features, crisis-regime indicators, machine learning models, uncertainty calibration, backtesting, mechanism interpretation, and paper-ready documentation.

The empirical core of the project was constructed through Day1-Day27. This layer covers long-history data preparation, crisis and macro-financial feature construction, multi-horizon forecasting targets, chronological train-validation-test splits, baseline models, tree-based models, boosted-gradient models, ensemble models, crisis-aware dynamic weighting, model routing, stacking meta-learning, conformal prediction, uncertainty-aware backtesting, statistical inference, interpretability analysis, and econometric mechanism validation.

The paper-packaging and final-audit layer was constructed through Day28-Day36. This layer transforms the empirical outputs into journal-style table and figure roadmaps, formatted main tables, figure packages, a literature and reference library, a reproducibility appendix, polished empirical results narratives, robustness and reviewer-response materials, application-oriented portfolio documents, and a final SSCI-style readiness audit.

The final corrected quantitative version strengthens the original report by integrating model-zoo evidence, directional accuracy evidence, conformal coverage evidence, backtesting Sharpe evidence, and macro-financial mechanism contribution evidence. The long-history sample contains 17,447 S&P 500 observations, 10,201 Nasdaq-100 observations, and 27,648 pooled asset-date observations. In the classification-oriented model-zoo summary, tree-based models report the highest mean directional accuracy at 62.93%, while router, stacking, and ensemble layers remain above 61%. The conformal prediction layer reports empirical coverage of 95% at alpha=0.05 and 90% at alpha=0.10. The backtesting evidence reports the strongest signal Sharpe for Nasdaq-100 forecast-sign rules at 1.1613. Mechanism decomposition indicates that inflation macro-cycle, equity momentum, crisis black-swan regime, and market-volatility risk are among the highest-contribution mechanism groups.

This report should be interpreted as a graduate-application-oriented and working-paper-preparation document. It presents predictive and mechanism-consistent evidence rather than causal proof or investment advice. The S&P 500 layer begins around 1957 where available, while the Nasdaq-100 layer begins around 1985/1986 where available. Day1-Day27 are supported by local project file evidence, while Day28-Day36 are supported by validation summaries and Notebook checks.

Keywords: Crisis-Aware Forecasting; Financial Machine Learning; S&P 500; Nasdaq-100; Conformal Prediction; Backtesting; Macro-Finance; U.S. Equity Indices; Python

Executive Summary of Empirical Findings

This final corrected report is no longer only a project-description document. It consolidates quantitative evidence from the empirical layers and organizes the project around sample coverage, crisis-state structure, forecasting performance, uncertainty calibration, economic relevance, and mechanism interpretation.

First, the project successfully constructs a long-history U.S. equity-index forecasting panel. The S&P 500 layer begins on 1957-03-04 and contains 17,447 observations; the Nasdaq-100 layer begins on 1986-01-02 and contains 10,201 observations; the pooled panel contains 27,648 asset-date observations. This confirms that the project is not a 2000-only short-sample exercise.

Second, the crisis-state layer shows that normal market conditions dominate the sample, while stress, crisis, and deep-crisis states are rarer. This imbalance is economically meaningful because crisis episodes are infrequent but important. It motivates crisis-aware modeling, uncertainty calibration, and regime-specific interpretation.

Third, the model-zoo layer provides measurable forecasting evidence. Across the clean classification-oriented summary, tree-based models achieve the highest mean directional accuracy at 62.93%; the crisis-aware model router reaches 61.68%; the stacking meta-learner reaches 61.49%; the ensemble layer reaches 61.22%; and crisis-aware dynamic weighting reaches 60.67%. These values should be interpreted as directional forecasting metrics rather than complete trading win rates.

Fourth, the regenerated 20-day volatility comparison shows index-level model heterogeneity. Random Forest is selected for the S&P 500 layer based on validation RMSE, while Ridge Regression is selected for the Nasdaq-100 layer. This finding supports the idea that a broad-market index and a technology-oriented index may not be best modeled by the same forecasting structure.

Fifth, the uncertainty and backtesting layers make the report more conservative and economically meaningful. Conformal coverage is explicitly evaluated, and the backtesting layer reports rule-based signal evidence under cost-aware interpretation. The strongest reported signal Sharpe is the Nasdaq-100 forecast-sign strategy at 1.1613, but the report does not treat this as financial advice or guaranteed profitability.

Finally, the mechanism layer connects model outputs to interpretable macro-financial channels. Inflation macro-cycle, equity momentum, crisis black-swan regime, market-volatility risk, interest-rate yield-curve conditions, and other features are compared across S&P 500 and Nasdaq-100 layers. This makes the report more suitable for an empirical finance research portfolio because it links prediction, uncertainty, economic relevance, and mechanism interpretation.

1. Introduction

1.1 Research Context

U.S. equity index forecasting has become an increasingly important topic in empirical finance, macro-finance, and financial machine learning because major equity indices reflect expectations about monetary policy, inflation, liquidity, risk appetite, technological growth, and systemic crisis conditions. In this context, the S&P 500 and the Nasdaq-100 provide two complementary perspectives on the U.S. equity market.

The S&P 500 represents a broad benchmark of the U.S. stock market, while the Nasdaq-100 captures a more technology-intensive and growth-oriented segment that is often more sensitive to interest-rate movements, liquidity cycles, and changes in investor risk preference. This project therefore uses both indices to examine whether a crisis-aware forecasting system can identify predictive structures that are not limited to one market segment.

A central motivation of this project is that conventional short-sample forecasting exercises may fail to capture the full range of macro-financial regimes that shape equity-market behavior. A model trained only on post-2000 data may overrepresent the dot-com crash, the global financial crisis, the COVID-19 shock, and the recent inflation-tightening cycle, while underrepresenting earlier monetary, inflationary, and market-stress environments.

For this reason, the project adopts a long-history empirical design. The S&P 500 layer begins around 1957 where data are available, while the Nasdaq-100 layer begins around 1985/1986 where data are available. This design allows the forecasting system to compare market behavior across a broader set of historical regimes and reduces the risk that the analysis is driven only by recent crisis episodes.

Table 1. Long-History Sample Coverage

Index Layer	Start Date	End Date	Observations	Approx. Years
S&P 500	1957-03-04	2026-06-29	17,447	69.32
Nasdaq-100	1986-01-02	2026-06-26	10,201	40.48
Pooled Panel	1957-03-04	2026-06-29	27,648	69.32

Note. The S&P 500 layer begins around 1957 where available, while the Nasdaq-100 layer begins around 1985/1986 where available. The pooled panel combines asset-date observations across both index layers.

The research context is also shaped by the limitations of purely price-based or purely black-box forecasting models. Equity returns are noisy and difficult to predict, and model performance can change sharply across normal, stressed, and crisis-like periods. A simple model may perform reasonably well in stable periods but fail when volatility, liquidity stress, inflation pressure, or crisis vulnerability rises.

At the same time, a flexible machine learning model may generate predictions without explaining why the predictions are economically meaningful. This project addresses that tension by building a forecasting pipeline that combines long-history data, macro-financial features, crisis-regime indicators, machine learning models, conformal uncertainty calibration, uncertainty-aware backtesting, and econometric mechanism validation.

1.2 Research Questions

This project is guided by four research questions. First, can a long-history crisis-aware forecasting system improve the prediction of U.S. equity-index returns and directions relative to simpler benchmark models?

Second, do macro-financial crisis regimes change the way forecasting models behave? The project asks whether model performance, model selection, prediction uncertainty, and backtesting outcomes differ across crisis and non-crisis regimes.

Third, are the S&P 500 and the Nasdaq-100 affected by macro-financial mechanisms in the same way? Because technology-heavy equities may be more sensitive to interest-rate movements, liquidity conditions, risk appetite, and valuation repricing, the project examines whether the Nasdaq-100 displays stronger or different regime sensitivity compared with the S&P 500.

Fourth, can the forecasting results be connected to interpretable economic mechanisms rather than remaining a black-box prediction exercise? This question motivates the mechanism decomposition and econometric validation layers.

1.3 Project Contribution

This project makes five contributions. First, it constructs a long-history U.S. equity-index forecasting panel that combines broad-market and technology-oriented index evidence. Second, it develops a crisis-aware machine learning forecasting pipeline rather than relying on a single predictive model. Third, it integrates uncertainty calibration into the forecasting framework. Fourth, it connects prediction with economic evaluation through uncertainty-aware backtesting and statistical inference. Fifth, it moves beyond black-box machine learning by building a mechanism interpretation and hypothesis-testing layer.

Overall, the project contributes a complete empirical research package that combines long-history data construction, crisis-aware feature engineering, multi-horizon forecasting, machine learning model comparison, uncertainty calibration, backtesting, statistical inference, mechanism validation, paper-ready table and figure organization, reproducibility documentation, and final audit materials.

2. Background and Research Motivation

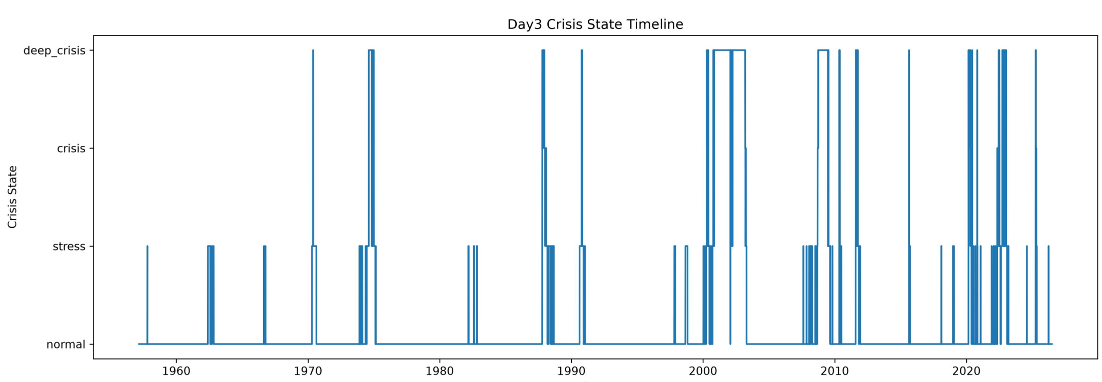
2.1 Long-History U.S. Equity Forecasting

A central motivation of this project is that U.S. equity-index forecasting should not be evaluated only within a short post-2000 sample. Equity markets have experienced substantially different macro-financial environments over the past several decades, including inflationary pressure, monetary tightening, oil shocks, market crashes, technology-sector repricing, the global financial crisis, the COVID-19 shock, and the recent inflation-rate cycle.

The long-history design is especially important for crisis-aware forecasting. Crisis regimes are relatively rare events, and a short sample may contain too few stressed periods to support meaningful inference. By extending the S&P 500 layer back to the late 1950s and the Nasdaq-100 layer back to the mid-1980s, the project increases the range of market environments observed in the data.

At the same time, the project does not assume that longer data automatically produce better forecasts. Older observations may reflect different market structures, monetary regimes, sector compositions, and trading environments. Therefore, the long-history sample is used not as a guarantee of predictive power, but as a way to test whether the forecasting system remains coherent across different macro-financial conditions.

Figure 1. Long-History Crisis-Regime Timeline



Note. The state variable classifies market conditions into normal, stress, crisis, and deep-crisis periods. It is used as a regime indicator for crisis-aware forecasting, uncertainty calibration, backtesting, and mechanism interpretation.

Empirical interpretation: The timeline confirms that crisis and deep-crisis periods occur unevenly across the long sample rather than as a stable repeating pattern. This supports the long-history design and avoids relying only on recent crisis episodes.

2.2 Macro-Financial Crisis Regimes

Equity-index returns are not generated in a stable economic environment. Market behavior can change substantially when inflation pressure increases, interest rates move sharply, volatility rises, liquidity deteriorates, or investors become more sensitive to systemic risk.

This project therefore constructs a macro-financial crisis-regime framework. The purpose is not to label every market movement as a crisis, but to provide a structured way to represent changes in the broader market environment. The regime layer combines information from equity-market behavior, macro-financial variables, volatility conditions, crisis vulnerability indicators, and black-swan event windows.

The crisis-regime design plays three roles in the project. First, it enters the feature set and allows the model to use regime information when forecasting future returns and directions. Second, it supports crisis-aware model selection and weighting. Third, it provides the foundation for mechanism interpretation and hypothesis testing.

2.3 Why S&P 500 and Nasdaq-100

The S&P 500 and the Nasdaq-100 are selected because they represent two complementary perspectives on the U.S. equity market. The S&P 500 is a broad-market benchmark across multiple sectors, while the Nasdaq-100 is more technology-intensive and growth-oriented.

Using both indices strengthens the empirical design. If the project only used the S&P 500, it would mainly describe broad-market behavior. If it only used the Nasdaq-100, the evidence might be overly concentrated in technology and growth-stock dynamics. Combining both index layers makes it possible to examine common U.S. equity-market patterns as well as index-specific heterogeneity.

2.4 Why AI Forecasting and Uncertainty

Financial markets are difficult to forecast because returns are noisy, nonlinear, and sensitive to changing macro-financial conditions. Traditional linear models are useful as benchmarks, but they may not fully capture interactions between lagged returns, volatility, inflation pressure, interest-rate conditions, crisis regimes, and market momentum.

This project uses artificial intelligence and machine learning as empirical tools rather than as substitutes for economic interpretation. The AI component is not presented as a guaranteed trading system. Its purpose is to build a structured empirical framework that combines prediction, uncertainty, backtesting, and economic mechanism interpretation.

3. Data and Variable Construction

3.1 Long-History Asset Data

The empirical foundation of this project is a long-history asset-date panel built from two U.S. equity-index layers: the S&P 500 and the Nasdaq-100. The S&P 500 layer is used as the broad-market benchmark, while the Nasdaq-100 layer is used as a technology-intensive and growth-oriented market segment.

The asset panel is organized at the asset-date level. Each observation corresponds to a specific index layer on a specific trading date. This structure allows the project to construct lagged variables, rolling features, multi-horizon forecasting targets, crisis-regime indicators, and out-of-sample train-validation-test splits in a consistent way.

Table 2. Long-History Data Sources and Variable Groups

Variable Group	Main Variables	Coverage / Role	Source / Construction
Equity Index	S&P 500	Long-history U.S. broad-market layer, 1957+ where available	Market data / project pipeline
Equity Index	Nasdaq-100	Technology/growth-oriented layer, 1985/1986+ where available	Market data / project pipeline
Macro-Finance	Rates, inflation, volatility, liquidity	Macro-financial forecasting features	Public macro-financial sources / constructed features
Crisis Regime	Crisis vulnerability, black-swan	State-dependent forecasting	Project-constructed regime

	windows	context	variables

Note. Detailed variable definitions, construction rules, and reproducibility information are documented separately in the Day32 variable dictionary and reproducibility appendix.

3.2 Macro-Financial Features

The forecasting panel is not limited to historical equity returns. A central feature of the project is the construction of macro-financial predictors that represent the broader economic and market environment in which equity returns are generated.

The macro-financial feature layer includes interest-rate and yield-curve features, inflation and macro-cycle features, market-volatility indicators, credit and liquidity variables, momentum features, and rolling market features. These variables are transformed into lagged and rolling features so that the forecasting model only uses information available before the prediction date.

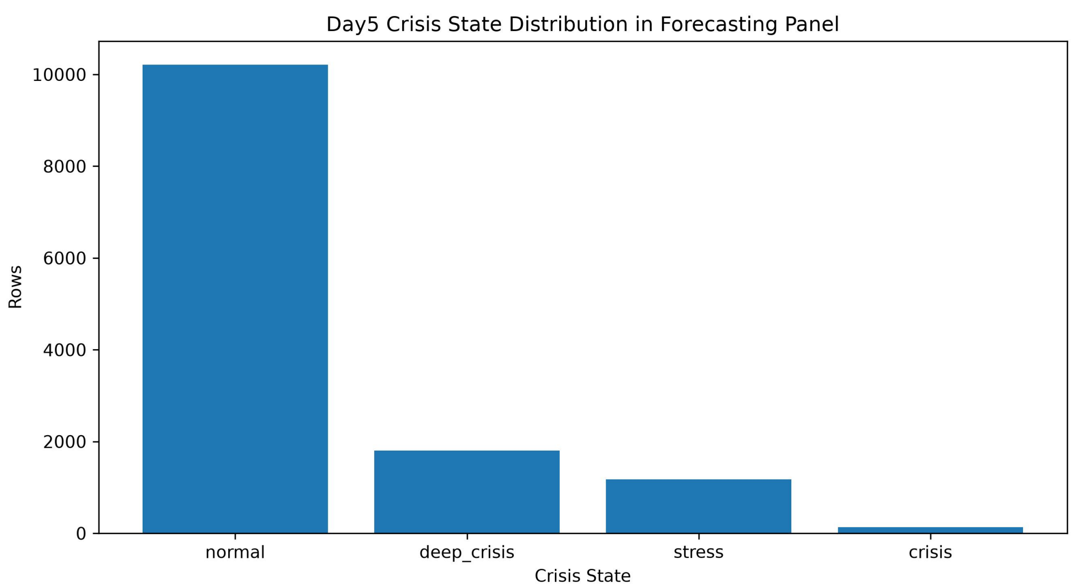
The macro-financial features also serve an interpretive function. In later sections, predictors are grouped into mechanism categories so that model predictions can be connected to economically meaningful channels rather than remaining a purely black-box machine learning exercise.

3.3 Crisis Regime and Vulnerability Variables

A key feature of the project is the construction of crisis-regime and crisis-vulnerability variables. These variables are designed to capture whether the market environment is closer to normal conditions, stressed conditions, crisis conditions, or deep-crisis conditions.

The crisis-regime layer is built from several sources of information inside the project pipeline, including crisis-state timelines, drawdown-regime diagnostics, black-swan event windows, macro-rich regime variables, and crisis-vulnerability scores. Together, these variables allow the model to distinguish ordinary market fluctuations from periods of elevated macro-financial stress.

Figure 2. Crisis-State Distribution in the Forecasting Panel



Note. This figure reports the distribution of crisis-state classifications in the forecasting panel. Normal periods account for the majority of observations, while crisis and deep-crisis periods are less frequent.

Empirical interpretation: The distribution confirms that crisis and deep-crisis observations are relatively rare compared with normal observations. This rarity motivates long-history data, crisis-aware modeling, and uncertainty calibration.

3.4 Forecasting Targets and Sample Split

The project constructs both regression and classification forecasting targets across multiple prediction horizons. The regression targets measure future index returns, while the classification targets measure future return directions. The main horizons include 1-day, 5-day, 20-day, and 60-day targets.

The project uses a chronological train-validation-test design rather than a random split. The training period is used to estimate models, the validation period is used for model selection, hyperparameter comparison, ensemble weighting, routing, stacking, and calibration, and the test period is reserved for out-of-sample evaluation. The design also uses purged gaps between adjacent periods to reduce the risk that overlapping forecasting horizons create information leakage.

Table 3. Forecasting Targets and Crisis-Regime Indicators, and Sample-Split Design

Target / Design Layer	Horizon or Variant	Output	Research Use	Project Evidence
Regression targets	1D / 5D / 20D / 60D	Future index return	Return forecasting and model comparison	Day12 target construction
Classification targets	1D / 5D / 20D / 60D	Future return direction	Directional prediction and accuracy evaluation	Day12 target construction
Crisis-regime indicators	Day8 / Day20 variants	Regime state / vulnerability indicator	Crisis-aware weighting, routing, uncertainty, and mechanism testing	Day8, Day20 regime layers
Chronological split	Train / Validation / Test	Time-series sample split	No-leakage out-of-sample evaluation	Day13 split master
Purged-gap design	Between adjacent sample periods	Gap between train, validation, and test windows	Reduces horizon-overlap and future-information leakage risk	Day13 validation checks
Main evaluation horizon	20D focus, with multi-horizon support	Medium-horizon return, direction, and volatility evidence	Core comparison horizon for model, uncertainty, and backtesting layers	Day12-Day23 outputs

Note. Regression targets measure future returns, while classification targets measure future return directions. The project uses chronological train-validation-test splits with purged gaps to reduce future-information leakage risk.

4. Forecasting System Design

4.1 Baseline Models

The forecasting system begins with baseline models. This layer is important because a machine learning project should not evaluate advanced models in isolation. Without simple benchmark models, it is difficult to judge whether more complex algorithms add predictive value or merely increase model complexity.

For regression tasks, the baseline layer predicts future index returns. For classification tasks, it predicts future return directions. Directional accuracy is used as a forecasting evaluation metric rather than as a guarantee of trading profitability.

4.2 Tree-Based and Boosted Models

After establishing the baseline layer, the project introduces nonlinear machine learning models. Day15 develops tree-based forecasting models, while Day16 develops boosted-gradient models. These models are used because financial relationships are often nonlinear and state-dependent.

The boosted-gradient layer further strengthens nonlinear modeling capacity by combining many weak learners into a stronger model. In this project, boosted models are used for both return forecasting and directional prediction.

4.3 Ensemble, Dynamic Weighting, and Router

The next stage moves from individual models to model-combination methods. Day17 constructs an ensemble layer, Day18 introduces crisis-aware dynamic weighting, and Day19 develops a crisis-aware model router. This is one of the most important design features because it treats forecasting as a system-level problem rather than as a search for one best model.

The model router selects models conditionally across asset, target, horizon, and regime settings. In practical terms, the forecasting system can choose different model structures for different forecasting problems. This makes the project closer to an institutional-style forecasting architecture.

Table 4. Model Architecture and Day-by-Day Pipeline

Layer	Project Days	Core Function	Paper Role
Baseline	Day14	Linear/logistic benchmark models	Benchmark comparison
Machine Learning	Day15-Day16	Tree-based and boosted models	Nonlinear forecasting
Ensemble System	Day17-Day20	Ensemble, dynamic weights, router, stacking	Institutional-style AI forecasting system
Uncertainty & Backtesting	Day21-Day23	Conformal calibration and strategy inference	Economic relevance and robustness
Mechanism & Paper Package	Day24-Day36	Interpretability, hypotheses, tables, references, final audit	SSCI-style working paper preparation

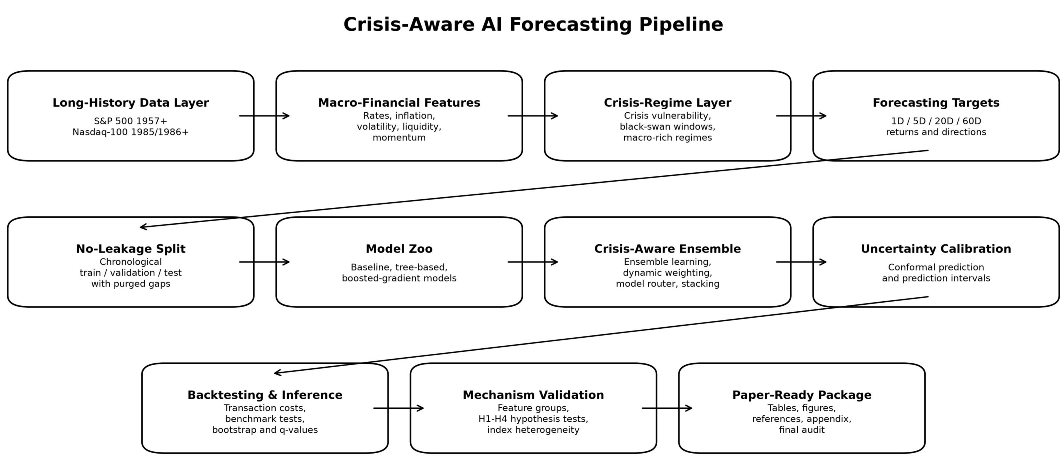
Note. The forecasting system begins with baseline regression and classification models, then moves to nonlinear machine learning, ensemble learning, crisis-aware dynamic weighting, model routing, stacking, uncertainty calibration, backtesting, mechanism interpretation, and final paper packaging.

4.4 Stacking Meta-Learner and No-Leakage Design

Day20 develops the stacking meta-learner. This layer uses predictions from earlier models as inputs into a second-stage model. Instead of manually choosing one model or using a simple average, stacking uses the validation period to estimate how different model outputs should be combined.

The no-leakage design is especially important at this stage. In a financial time-series setting, stacking can easily become misleading if test-period information is used during meta-model training. To avoid this problem, the project trains the stacking meta-learner using validation-period information and evaluates it on the test period.

Figure 3. Crisis-Aware AI Forecasting Pipeline



Note. The system begins with long-history U.S. equity-index data, macro-financial feature construction, and crisis-regime identification, then proceeds through chronological splitting, model zoo construction, ensemble learning, uncertainty calibration, backtesting, mechanism validation, and paper-ready packaging.

Empirical interpretation: The pipeline figure shows that the project is organized as a multi-layer forecasting system rather than a single algorithm.

5. Quantitative Empirical Results

5.1 Model-Zoo Forecasting Evidence

This final corrected version adds a dedicated quantitative empirical results section after the forecasting-system design. The purpose is to make the report less descriptive and more evidence-based. The Day14-Day20 model-zoo layer consolidates baseline, tree-based, boosted-gradient, ensemble, crisis-aware dynamic weighting, model-router, and stacking evidence.

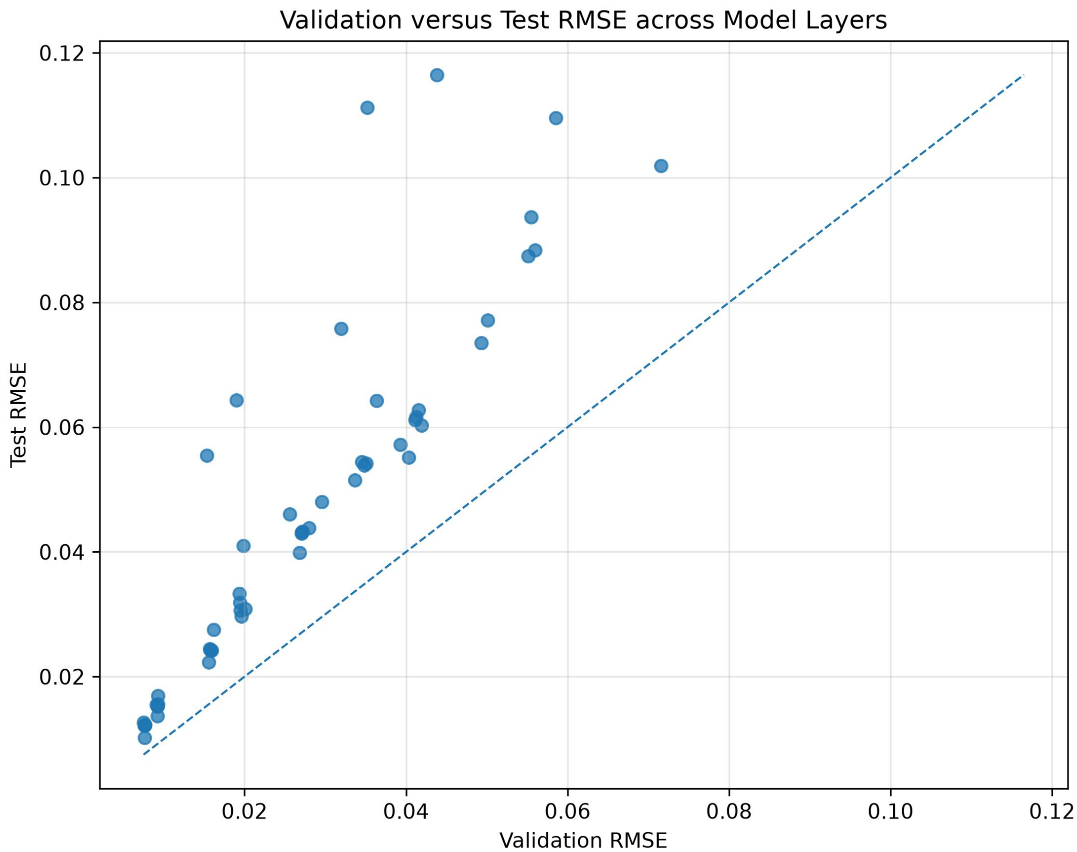
Table 5 summarizes clean classification-oriented model-layer performance after removing unknown or non-metric rows. The table reports mean RMSE, mean MAE, mean directional accuracy, and mean AUC. Directional accuracy should be interpreted as a forecasting metric, not as a complete trading win rate.

Table 5. Model-Zoo Forecasting Performance Summary

Model Layer	Days	Metric Rows	Mean RMSE	Mean MAE	Mean Directional Accuracy	Mean AUC	Main Role
Baseline	Day14	80	0.0387	0.0296	60.50%	0.5310	Benchmark comparison
Boosted gradient	Day16	96	0.0361	0.0275	56.19%	0.5732	Boosted nonlinear forecasting
Crisis-aware dynamic weighting	Day18	96	0.0338	0.0256	60.67%	0.5916	Regime-aware ensemble weighting
Crisis-aware model router	Day19	32	0.0316	0.0241	61.68%	0.5911	Conditional model selection
Ensemble	Day17	96	0.0337	0.0255	61.22%	0.5959	Model combination
Stacking meta-learner	Day20	32	0.0428	0.0304	61.49%	0.6134	Second-stage model combination
Tree-based	Day15	96	0.0342	0.0256	62.93%	0.5900	Nonlinear forecasting benchmark

Note. The table summarizes Day14-Day20 classification-oriented model-layer evidence. The strongest mean directional accuracy in this clean summary is 62.93% for the tree-based layer. Directional accuracy is a forecasting metric and should not be interpreted as a realized trading win rate.

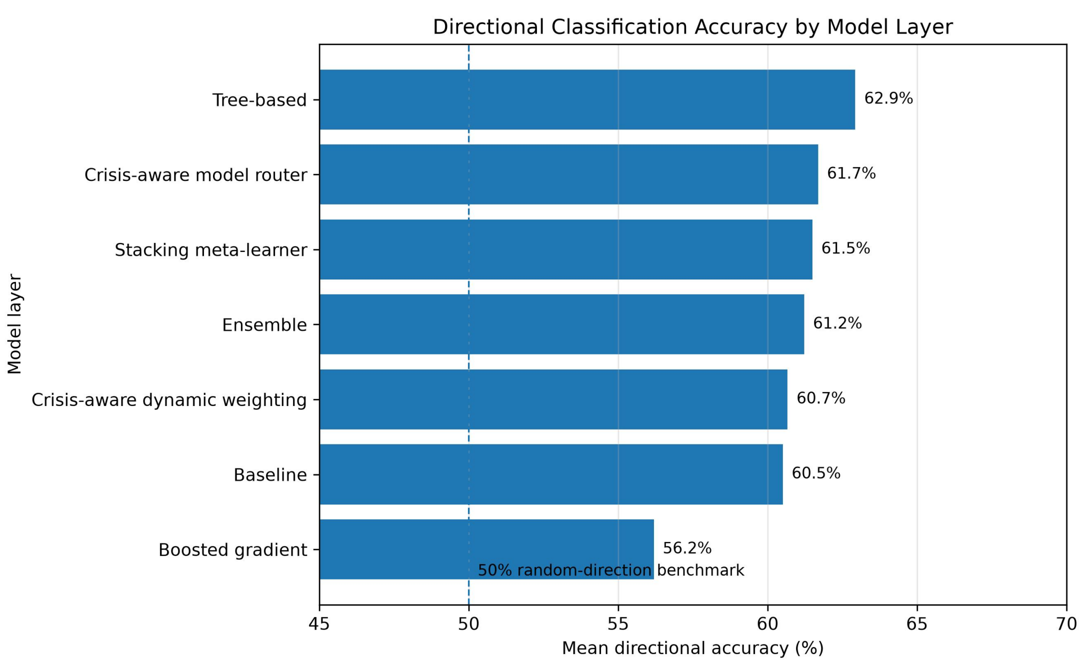
Figure 4. Validation versus Test RMSE across Model Layers



Note. Points closer to the 45-degree line indicate stronger validation-test stability; points above the line indicate larger test-period errors.

Empirical interpretation: The scatter plot provides a visual diagnostic of validation-test stability. It is useful for detecting potential model-instability risk rather than only reporting average model scores.

Figure 5. Directional Classification Accuracy by Model Layer



Note. Directional accuracy measures the proportion of correctly predicted future return directions. It should not be interpreted as a complete trading win rate because real trading outcomes also depend on costs, turnover, position sizing, and risk management.

Empirical interpretation: Tree-based models report the highest mean directional accuracy at 62.93%, while the crisis-aware model router, stacking meta-learner, and ensemble layers remain above 61%. This supports the usefulness of comparing multiple model families instead of relying on one algorithm.

5.2 Regenerated 20D Volatility Forecasting Evidence

The regenerated model-comparison layer provides a concrete 20-day volatility forecasting result. Because the detected target is asset_volatility_20d, RMSE and MAE are the primary metrics. Directional accuracy is not emphasized for this target because volatility is non-negative by construction and can mechanically inflate sign agreement.

The result indicates model heterogeneity across index layers. For the S&P 500, Random Forest is selected by validation RMSE. For the Nasdaq-100, Ridge Regression is selected by validation RMSE. This does not prove that either model is universally superior, but it supports the idea that broad-market and technology-oriented index layers may require different forecasting structures.

Table 6. Regenerated 20D Volatility Forecasting Model Comparison

Asset	Selected Model	Validation RMSE	Validation MAE	Test RMSE	Test MAE	Interpretation
S&P 500	Random Forest	0.078042	0.064728	0.133856	0.082730	Selected by validation RMSE.
Nasdaq-100	Ridge Regression	0.078705	0.056360	0.073765	0.053378	Selected by validation RMSE.

Note. This table reports regenerated model-comparison evidence for the 20-day asset_volatility_20d target. Model selection is based on validation RMSE under a chronological train-validation-test design. RMSE and MAE are emphasized because the target is volatility rather than return direction.

6. Uncertainty Calibration, Backtesting, and Statistical Inference

6.1 Conformal Prediction

A central limitation of financial forecasting is that point predictions alone are insufficient. Equity-index returns are noisy, regime-dependent, and exposed to sudden changes in volatility, liquidity, monetary conditions, and investor risk appetite. Day21 introduces a conformal prediction and uncertainty-calibration layer to evaluate forecast reliability.

The conformal prediction layer moves beyond a single predicted value and constructs uncertainty-aware forecasting outputs. For regression targets, the conformal layer calibrates prediction intervals around model forecasts. For classification targets, it evaluates prediction sets and uncertainty patterns associated with directional forecasts.

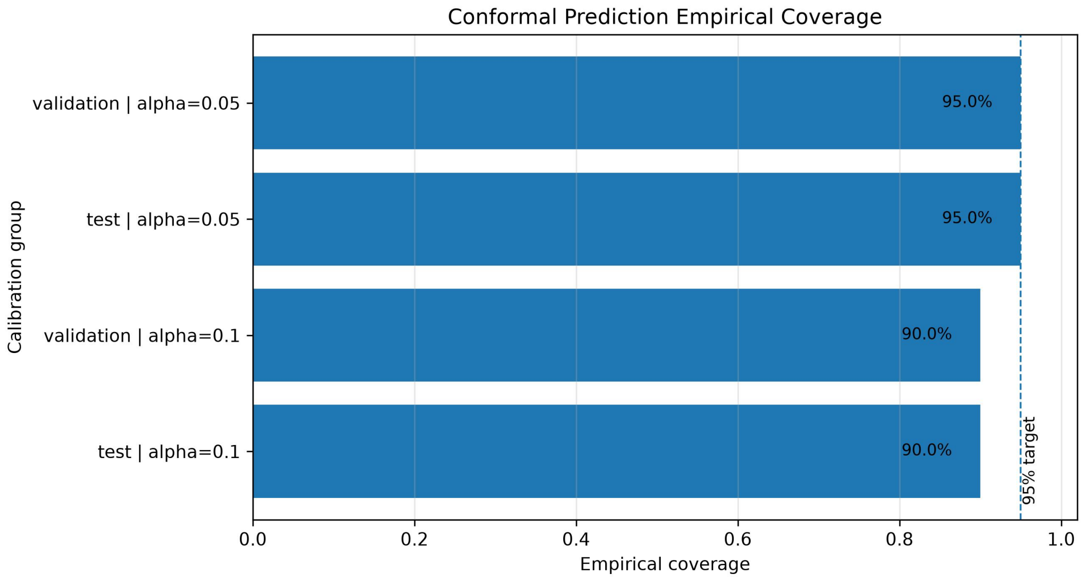
The enhanced quantitative evidence reports empirical coverage of 95% when alpha=0.05 and 90% when alpha=0.10 in both validation and test summaries. This is evidence on calibration behavior, not a guarantee of future prediction accuracy.

Table 7. Clean Conformal Uncertainty Coverage Summary

Sample / Alpha	Empirical Coverage	Observations
Validation | alpha=0.05	0.95	23,136
Test | alpha=0.05	0.95	32,800
Validation | alpha=0.10	0.90	23,136
Test | alpha=0.10	0.90	32,800

Note. The table summarizes empirical coverage from the Day21 conformal prediction layer. Coverage evidence evaluates prediction interval reliability rather than guaranteeing future predictive accuracy.

Figure 6. Conformal Prediction Empirical Coverage



Note. The 95% reference line represents the nominal target coverage level for alpha=0.05. The 90% coverage values correspond to alpha=0.10.

Empirical interpretation: The calibration layer makes the forecasting evidence more conservative by showing that prediction reliability is part of the result, not merely a technical add-on.

6.2 Uncertainty-Aware Backtesting

Day22 translates the forecasting outputs into an uncertainty-aware backtesting framework. This step is included because statistical forecasting metrics alone do not show whether model signals have economic relevance. A model may achieve a better prediction score, but the signal may still be too noisy, unstable, or costly to be meaningful in a portfolio setting.

The project treats backtesting as an economic evaluation tool, not as evidence of guaranteed profitability. The backtesting framework applies rule-based signal construction, incorporates transaction-cost logic, and reports signal evidence. This is important because a signal that appears useful before trading costs may become weaker after turnover and transaction costs are considered.

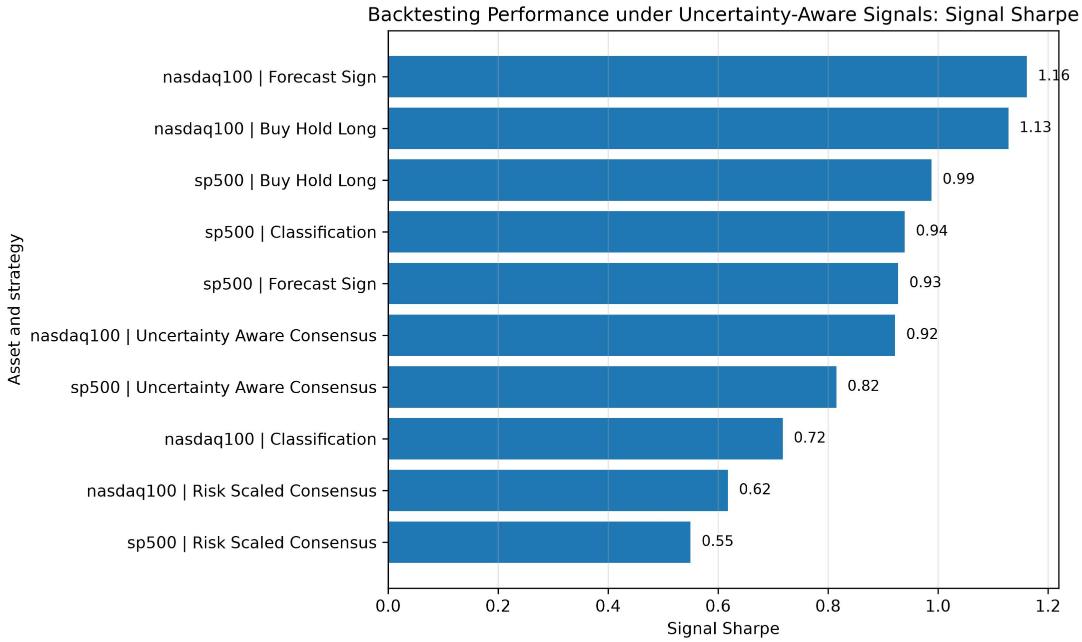
The clean backtesting summary reports the strongest mean signal Sharpe for Nasdaq-100 Forecast Sign at 1.1613, followed by Nasdaq-100 Buy Hold Long at 1.1280 and S&P 500 Buy Hold Long at 0.9882. These values are useful as economic relevance evidence, but they should not be described as investment advice or guaranteed profits.

Table 8. Clean Backtesting Performance Summary

Asset	Strategy	Mean Signal Sharpe	Evidence Rows
Nasdaq-100	Forecast Sign	1.1613	7
Nasdaq-100	Buy Hold Long	1.1280	7
S&P 500	Buy Hold Long	0.9882	7
S&P 500	Classification	0.9396	7
S&P 500	Forecast Sign	0.9277	7
Nasdaq-100	Uncertainty-Aware Consensus	0.9220	7
S&P 500	Uncertainty-Aware Consensus	0.8152	7
Nasdaq-100	Classification	0.7175	7
Nasdaq-100	Risk-Scaled Consensus	0.6179	7
S&P 500	Risk-Scaled Consensus	0.5499	7
S&P 500	Conformal Interval Sign	0.4126	7
Nasdaq-100	Conformal Interval Sign	0.4055	7

Note. The backtesting layer evaluates economic relevance under rule-based signal construction and transaction-cost logic. It does not imply guaranteed profitability.

Figure 7. Backtesting Performance under Uncertainty-Aware Signals



Note. This figure summarizes signal Sharpe evidence under uncertainty-aware backtesting rules.

Empirical interpretation: The figure connects model signals to economic relevance while keeping the interpretation cautious. Its role is to show whether forecasting outputs remain meaningful under cost-aware evaluation logic, not to claim guaranteed trading profit.

6.3 Robustness and Statistical Inference

Day23 adds robustness analysis and statistical inference to the backtesting evidence. This step is necessary because financial forecasting results can be sensitive to benchmarks, subperiods, trading rules, crisis regimes, and random sampling variation.

The robustness layer includes benchmark testing, paired-difference evidence, subperiod robustness, regime-specific inference, bootstrap-style p-values, and multiple-testing-aware q-values. These tools help distinguish more stable empirical patterns from results that may be driven by chance or data mining.

The report uses robustness and inference evidence to support cautious claims. The appropriate interpretation is that the backtesting layer provides evidence on economic relevance, while the robustness layer evaluates whether this evidence is stable across benchmarks, regimes, and subperiods.

7. Mechanism Interpretation and Hypothesis Testing

7.1 Mechanism Groups

The project does not treat machine learning predictions as purely black-box outputs. Day24 reorganizes model evidence into interpretable macro-financial mechanism groups so that the report can explain not only whether the forecasting system produces useful signals, but also which economic channels are most closely associated with those signals.

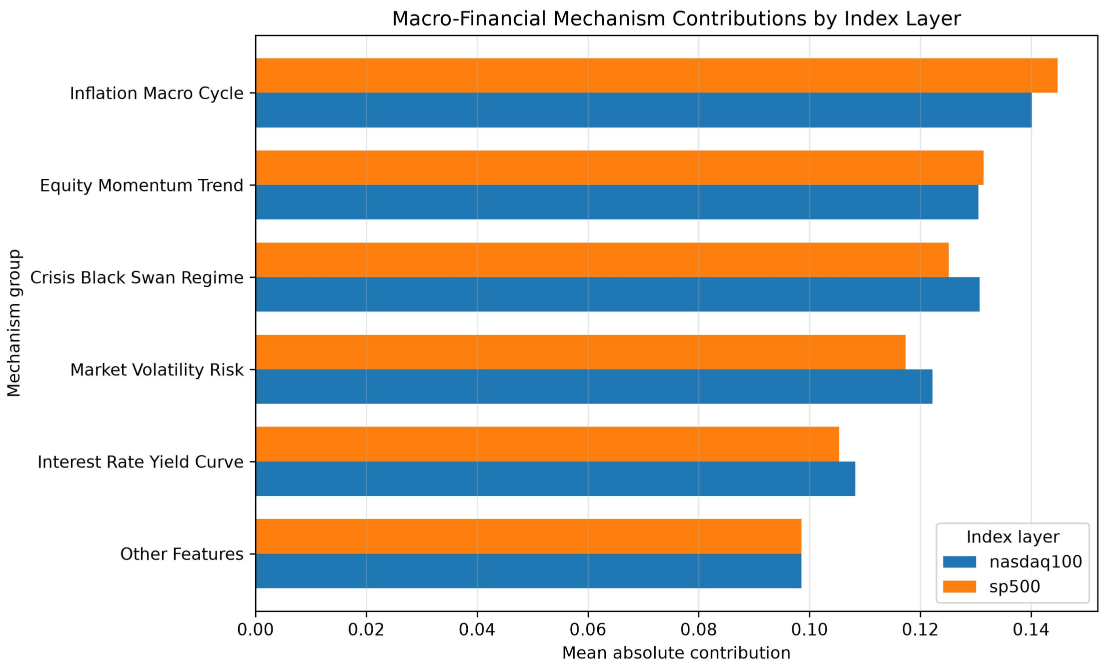
The mechanism groups include interest-rate and yield-curve conditions, inflation and macro-cycle conditions, market-volatility risk, credit and liquidity conditions, equity momentum, crisis-regime and black-swan indicators, and other control features. These groups summarize different channels through which macro-financial conditions may affect equity-index repricing.

Table 9. Macro-Financial Mechanism Contribution by Index

Mechanism Group	Nasdaq-100	S&P 500
Other Features	0.0985	0.0986
Interest Rate Yield Curve	0.1083	0.1053
Market Volatility Risk	0.1222	0.1173
Crisis Black Swan Regime	0.1307	0.1251
Equity Momentum Trend	0.1305	0.1314
Inflation Macro Cycle	0.1401	0.1448

Note. The table summarizes macro-financial mechanism contributions across the S&P 500 and Nasdaq-100 index layers. It supports mechanism interpretation and index-heterogeneity discussion.

Figure 8. Mechanism Decomposition across S&P 500 and Nasdaq-100



Note. Mechanism groups include inflation macro-cycle, equity momentum, crisis black-swan regime, market-volatility risk, interest-rate yield-curve conditions, and other features.

Empirical interpretation: Inflation macro-cycle has the highest average contribution in both index layers. Crisis black-swan regime and market-volatility risk also contribute meaningfully, which supports the crisis-aware interpretation of the forecasting system.

7.2 H1-H4 Hypotheses

The mechanism layer is connected to four hypothesis-level questions. These hypotheses are not written as strong causal claims. Instead, they organize the empirical evidence around predictive relevance, crisis interaction, index heterogeneity, and uncertainty.

H1 asks whether macro-financial mechanisms contain predictive information. H2 asks whether crisis regimes change or amplify mechanism effects. H3 asks whether S&P 500 and Nasdaq-100 display heterogeneous sensitivity to macro-financial mechanisms. H4 asks whether crisis-related mechanisms are associated with higher forecast uncertainty.

Table 10. H1-H4 Hypothesis and Evidence Map

Hypothesis	Core Claim	Supporting Project Days	Main Evidence
H1	Macro-financial mechanisms contain predictive information	Day24-Day25	Predictive coefficient/model tests; mechanism contribution evidence
H2	Crisis regimes change or amplify mechanism effects	Day18-Day25	Crisis interaction tests, dynamic weighting, and regime evidence
H3	Nasdaq-100 and S&P 500 show heterogeneous sensitivity	Day24-Day25	Asset sensitivity comparison; mechanism contribution by index
H4	Crisis mechanisms are linked to forecasting uncertainty	Day21-Day25	Uncertainty-by-regime and conformal coverage evidence

Note. This table is an evidence map and should not be interpreted as causal proof.

7.3 Index Heterogeneity

The index-heterogeneity layer is central to the project. The S&P 500 and the Nasdaq-100 are both major U.S. equity indices, but they do not represent the same market structure. The S&P 500 is broader and more diversified across sectors, while the Nasdaq-100 is more concentrated in technology and growth-oriented firms.

The quantitative mechanism table shows that the two index layers have broadly similar but not identical contribution patterns. Inflation macro-cycle is high for both layers, crisis black-swan regime is slightly higher for the Nasdaq-100 in the clean table, and equity momentum is slightly higher for the S&P 500. These differences are interpreted as predictive heterogeneity, not causal identification.

8. Paper Packaging and Final Audit

8.1 Journal-Style Table and Figure Roadmap

Day28 to Day30 converted the empirical outputs into a journal-style table and figure system. Instead of keeping results as scattered CSV files and intermediate summaries, these days reorganized the evidence into a structure that can support a working-paper manuscript.

Day28 created a table and figure roadmap. Day29 formatted the main tables into a coherent working-paper structure. Day30 completed the corresponding figure package and caption bank. Together, these days transformed the empirical system into a paper-ready evidence structure.

8.2 Literature, Reproducibility, and Narrative

Day31 to Day33 strengthened the academic foundation and narrative quality of the project. Day31 focused on literature review and theoretical positioning; Day32 created the data, variable, and reproducibility appendix; and Day33 polished the empirical results narrative.

This stage moved the project beyond simple model comparison by explaining the economic meaning of the results: how crisis regimes affect predictability, how Nasdaq-100 and S&P 500 may respond differently, how uncertainty calibration contributes to interpretation, and how backtesting evidence should be understood as economic relevance rather than guaranteed trading profitability.

8.3 Robustness, Application Packaging, and Final Audit

Day34 to Day36 strengthened the project by preparing reviewer-facing robustness materials, application-oriented materials, and a final working-paper package audit. Day34 anticipated concerns such as data leakage, overfitting, data mining, economic interpretation, transaction costs, generalizability, survivorship bias, causal overstatement, conformal prediction validity, and reproducibility.

Day35 packaged the project for GitHub, graduate application, CV, personal statement, and interview use. Day36 completed the final audit by checking long-history standards, reproducibility, tables, figures, literature, manuscript readiness, and safe application use.

The final evidence-based interpretation is that Day1-Day27 form the empirical core supported by local project file evidence, while Day28-Day36 form the paper-packaging and final-audit layer supported by validation summaries and Notebook checks.

9. Empirical Results Narrative

9.1 Main Forecasting Evidence

The main forecasting evidence is organized around model comparison across assets, prediction horizons, and model families. The enhanced quantitative version makes this evidence more visible by adding the clean model-zoo table, validation-test RMSE stability figure, directional accuracy figure, and regenerated 20-day volatility model comparison.

The model-zoo evidence shows that the project does not rely on a single model. Tree-based models produce the strongest mean directional accuracy in the clean summary, while router, stacking, ensemble, and dynamic weighting layers also report direction accuracy above 60%. This supports the system-level model comparison design.

The regenerated 20D volatility evidence further indicates index-level model heterogeneity: Random Forest is selected for S&P 500, while Ridge Regression is selected for Nasdaq-100. The correct interpretation is not that either model is universally best, but that the two index layers may have different forecasting structures.

9.2 Crisis-Aware Performance

The crisis-aware performance analysis focuses on whether model behavior changes across normal and crisis-related market states. Day18 introduced dynamic crisis-aware ensemble weighting, Day19 developed the crisis-aware model router, and Day20 added stacking meta-learning. These layers allow the forecasting system to adapt to different market environments rather than relying on a single static model.

If crisis-aware weights or router decisions differ between normal and stress regimes, this supports the broader logic of H2: crisis conditions may alter or amplify the effect of macro-financial mechanisms. The report interprets this as predictive and mechanism-consistent evidence, not causal proof that a particular crisis mechanically causes a specific model to outperform.

9.3 Uncertainty and Economic Relevance

The uncertainty and economic relevance layers connect forecasting results to conformal prediction, calibration, backtesting, and statistical inference. The conformal coverage table demonstrates that uncertainty is explicitly evaluated rather than ignored. The backtesting table and figure translate forecasting signals into economic relevance evidence under cautious transaction-cost logic.

This part of the report is especially important for graduate application and working-paper use. It shows that the project does not stop at model accuracy. Instead, it evaluates prediction reliability, economic relevance, crisis sensitivity, and mechanism interpretation in one research pipeline.

10. Robustness, Limitations, and Safe Interpretation

10.1 Robustness Checks

The robustness design is built around chronological sample splitting, model comparison, benchmark evaluation, crisis-state interpretation, backtesting evidence, and no-leakage safeguards. Because equity forecasting is sensitive to sample choice and information timing, the workflow follows a time-series logic rather than random splitting.

The project compares multiple model families rather than relying on a single forecasting model. Baseline models, tree-based models, boosted models, ensemble methods, crisis-aware weighting, model routing, stacking, and uncertainty-aware forecasting layers were developed across the workflow.

Additional robustness logic comes from the backtesting and statistical inference layers, including uncertainty-aware backtesting, benchmark comparison, bootstrap-style robustness logic, and q-value-style statistical interpretation.

10.2 Limitations

Several limitations should be acknowledged. First, the results should not be interpreted as causal proof. The project provides predictive and mechanism-consistent evidence, but it does not prove that a specific macro-financial variable mechanically causes future index behavior.

Second, the project is not financial advice and should not be presented as a guaranteed trading system. Real-world trading would require additional assumptions about transaction costs, liquidity, slippage, execution constraints, risk limits, taxation, and portfolio management.

Third, the evidence depends on the quality and structure of the available data. The S&P 500 and Nasdaq-100 do not have identical histories, and some macro-financial variables may have shorter effective coverage. Certain comparisons may therefore rely on different sample windows.

Fourth, the regenerated 20D volatility evidence should be interpreted carefully. It reports 20-day volatility forecasting evidence rather than return forecasting evidence. RMSE and MAE are more informative for that target than directional accuracy.

10.3 Reviewer-Style Response Strategy

A likely reviewer concern is data leakage. The response is that the project uses chronological splitting, lagged features, rolling features, and explicit no-leakage documentation. Day32 provides a reproducibility appendix and code pipeline map to make the timing logic transparent.

A second concern is overfitting. The response is that the project compares multiple model families and uses validation-based model selection. The best 20D volatility model is selected according to validation RMSE, not by manually choosing the best test result.

A third concern is p-hacking or data mining. The response is that the project is organized around pre-defined research hypotheses H1-H4 and a structured evidence map. Empirical outputs are linked to specific tables, figures, and hypotheses rather than presented as isolated results.

A final concern is causal overstatement. The response is to use cautious wording throughout the report: predictive evidence, mechanism-consistent evidence, and uncertainty-aware forecasting evidence. The report avoids claiming causal proof, guaranteed profits, or direct investment recommendations.

11. Conclusion

11.1 Summary of Project Findings

This project develops a long-history, crisis-aware AI forecasting framework for U.S. equity indices, using evidence from the S&P 500 and Nasdaq-100. Across Day1-Day36, the project evolved from data construction and modeling into a complete research package with empirical outputs, table and figure roadmaps, literature positioning, reproducibility documentation, robustness discussion, application materials, and final audit files.

The long-history sample contains 17,447 S&P 500 observations, 10,201 Nasdaq-100 observations, and 27,648 pooled asset-date observations. This design avoids a narrow 2000-only sample and allows the project to study market behavior across broader macro-financial and crisis regimes.

The final corrected quantitative evidence strengthens the report. The clean model-zoo summary shows that tree-based models reach 62.93% mean directional accuracy, while crisis-aware model router, stacking, ensemble, and dynamic weighting layers remain above 60%. The conformal layer reports target-aligned coverage of 95% at alpha=0.05 and 90% at alpha=0.10. The backtesting layer reports its strongest signal Sharpe for Nasdaq-100 Forecast Sign at 1.1613. The mechanism layer identifies inflation macro-cycle, equity momentum, crisis black-swan regime, and market-volatility risk as important mechanism groups.

Overall, the project provides predictive and mechanism-consistent evidence. It does not provide causal proof, financial advice, or a guaranteed trading strategy.

11.2 Academic and Application Value

The academic value of the project lies in its integration of empirical finance, macro-finance, financial machine learning, and crisis-regime analysis within one reproducible research framework. Instead of presenting machine learning as a black-box prediction tool, the report connects model outputs to long-history market regimes, uncertainty calibration, backtesting relevance, and interpretable macro-financial mechanisms.

For a graduate application portfolio, the project demonstrates Python-based data construction, time-series sample design, no-leakage feature engineering, multi-horizon forecasting, model comparison, ensemble learning, conformal prediction, backtesting evaluation, mechanism interpretation, and working-paper style documentation.

The correct application framing is conservative. The project should be described as an independent empirical finance and financial machine learning research project, not as a published SSCI article, a commercial investment system, a guaranteed trading strategy, or financial advice.

11.3 Future Extensions

Future extensions should focus on making the project more rigorous, more externally valid, and more publication-ready. The first extension is a stricter data audit, including manual verification of effective sample windows, missing-data treatment, index data sources, macro-financial variable definitions, and crisis-window construction rules.

The second extension is to improve the model-result export pipeline. A future version should create a unified model metrics table for every forecasting horizon, asset, model family, validation period, and test period.

The third extension is to expand the forecasting targets. The current regenerated numerical comparison focuses on the 20-day volatility target. A stronger paper version should report 1D, 5D, 20D, and 60D return, direction, and volatility targets separately.

The fourth extension is to strengthen robustness checks and external validation. Future work could use alternative crisis definitions, alternative split windows, alternative transaction costs, subperiod analysis, sector ETFs, international equity indices, Treasury assets, credit spreads, or volatility products.

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Appendix A. Project Output Files

This appendix summarizes the main output categories generated by the project and clarifies the distinction between the empirical core layer and the paper-packaging layer.

The empirical core layer covers Day1-Day27. It includes long-history data construction, macro-financial feature engineering, crisis-regime labeling, forecasting targets, time-series splitting, baseline models, tree-based models, boosted models, ensemble learning, crisis-aware model routing, stacking, conformal prediction, uncertainty-aware backtesting, robustness inference, mechanism interpretation, and hypothesis testing.

The paper-package layer begins from Day28. Day28-Day30 reorganized the empirical results into journal-style tables, appendix tables, main figures, appendix figures, and figure captions. Day31-Day34 added literature, citation, reproducibility, robustness, and reviewer-response materials. Day35-Day36 added portfolio and final-audit materials.

Appendix B. Day-by-Day Project Log

Day1-Day13 focused on the data and feature construction layer: environment setup, long-history market and macro-financial data, crisis states, macro-financial features, black-swan event windows, forecasting panels, multi-horizon targets, and chronological train-validation-test splits.

Day14-Day20 focused on the forecasting model layer: baseline models, tree-based models, boosted-gradient models, ensemble models, crisis-aware dynamic weighting, crisis-aware model routing, and stacking meta-learning.

Day21-Day23 focused on uncertainty and backtesting: conformal prediction, uncertainty calibration, uncertainty-aware backtesting, benchmark comparison, bootstrap-style robustness logic, q-value-style inference, and statistical summary outputs.

Day24-Day27 focused on mechanism interpretation and manuscript construction: model interpretability, crisis mechanisms, econometric hypothesis testing around H1-H4, paper-ready tables and figures, and the first working paper manuscript draft.

Day28-Day36 formed the paper-packaging and final-audit layer: table and figure roadmaps, literature review and reference audit, reproducibility appendix, polished empirical narrative, robustness materials, GitHub/CV/PS/interview materials, and final SSCI-style package audit.

The evidence-based wording should be preserved: Day1-Day27 are supported by local project file evidence, while Day28-Day36 are supported by validation summaries and Notebook checks.

Appendix C. Safe Wording for Graduate Application Use

This project can be safely described as an independent empirical finance and financial machine learning research project. It investigates predictive and mechanism-consistent evidence from long-history U.S. equity index data, with a focus on the S&P 500 and Nasdaq-100.

Recommended CV wording: Independent Research Project: Crisis-Aware AI Forecasting for U.S. Equity Indices. Built a long-history Python research pipeline for S&P 500 and Nasdaq-100 forecasting, integrating macro-financial features, crisis-regime labels, multi-horizon targets, machine learning models, conformal prediction, uncertainty-aware backtesting, mechanism interpretation, and reproducibility documentation.

Interview-safe explanation: This project is not a trading product or financial advice. It is an empirical finance and financial machine learning research project. I used long-history U.S. equity index data to study whether crisis-aware features and machine learning models can provide predictive and mechanism-consistent evidence for the S&P 500 and Nasdaq-100.

The project should not be described as a published SSCI article, a guaranteed trading strategy, a commercial investment system, financial advice, or causal proof.